Appendix C: IEEE 830 Template

Software Requirements Specification

For
TUL Mobility Semester Quali Database (TMSQD)

Version 1.4

Prepared by Bartosz Mużdżak

12.06.2026

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Revision History

Name	Date	Reason for Changes	Version
Bartosz Mużdzak	27.04.2026	«”Initial version” or actual reason»	V. 1.0
Bartosz Mużdzak	08.05.2026	“First” draft after intro	V1.1
Bartosz Mużdżak	27.05.2026	Draft with implementation of first UML diagram	V1.2
Bartosz Mużdżak	27.05.2026	Draft with implementation of UML class-diagram	V1.3
Bartosz Mużdżak	12.06.2026	Final draft	V1.4

1. Introduction

1.1. Purpose

The purpose of this document is to define the comprehensive software requirements for the **TUL Mobility Semester Quality Database (TMSQD)**, version 1.0. This system is designed to improve recruitment and calculation processes, enable to students see, how many of them are going to the particular university, and mobility agreements within the TUL and Erasmus+ (Technical University of Lodz) ecosystem.

1.2. Document Conventions

This document follows the IEEE 830 standard. Requirement priorities are categorized as High, Medium, and Low. Every functional requirement follows the "The system shall..." format to ensure clarity and verifiability.

1.3. Intended Audience and Reading Suggestions

This document is intended for:

- **Developers:** To implement the logic and database schema.
- **Project Managers:** To monitor progress and scope.
- **Users (Staff/Students):** To understand the system's capabilities.
- **Testers:** To create test cases based on the functional specifications.

1.4. Product Scope

TMSQD is a centralized database system aimed at automating the qualification and validation process for student mobility semesters. It will handle course equivalencies, ECTS point tracking, and quality assurance metrics to ensure international standards are met as well as allowing students to check, where their peers are going, improving communications and potential cooperation.

1.5. References

TBU

2. Overall Description

2.1. Product Perspective

TMSQD acts as a standalone module that interacts with the university's main student information system. It focuses specifically on the "Quality" aspect of mobility, ensuring that academic standards are consistent across different institutions.

2.2. Product Features

- **Students:** Can view available mobility programs and track their qualification status and view general table of each student's destination.

- **Academic Coordinators and Mobility Division Staff:** Responsible for verifying course quality, approving learning agreements, and providing necessary data to calculate points for each individual.

- **Administrators:** Manage system access, database backups, and global configuration.

2.3. User Classes and Characteristics

The system distinguishes between three primary user classes: Students, Academic Coordinators, and System Administrators. Each class varies in access levels, ranging from students who only view data to administrators who manage the entire database infrastructure.

2.4. Operating Environment

The software will operate as a web-based application hosted on a central university server running Linux (Ubuntu 22.04 LTS). It must be compatible with all modern web browsers and coexist with the existing TUL student information ecosystem.

2.5. Design and Implementation Constraints

The developers must adhere to the university's data protection policies and GDPR regulations regarding sensitive student information. Implementation is restricted to using a relational SQL database and the React framework to ensure long-term maintenance compatibility by the university IT staff.

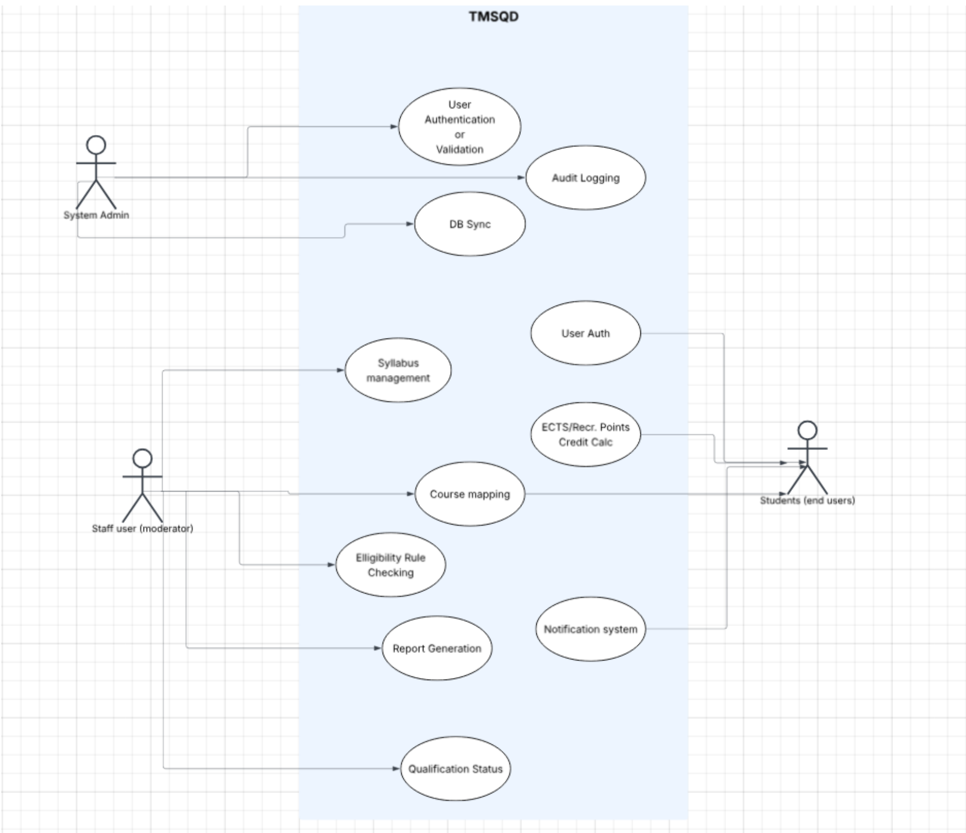
2.6. User Documentation

A comprehensive digital User Manual will be provided in PDF format, covering all system functionalities for both students and staff. Additionally, an integrated "On-line Help" tooltip system will be implemented within the GUI to assist users in real-time.

2.7. Assumptions and Dependencies

It is assumed that the university's central API will provide consistent and uninterrupted access to the student record database. The project also depends on the availability of the partner universities' course catalogues to populate the quality verification module.

3. System Features



3.1. Student Eligibility Verification

This feature automatically evaluates whether a student meets the necessary academic and ECTS requirements for a specific mobility program. It provides instant feedback to both the student and the coordinator regarding qualification status.

3.2. Course Equivalency Mapping

This module allows users to compare and link courses from foreign universities with the local curriculum at TUL. It ensures that the academic quality and credit weight of the mobility semester are fully recognized.

4. External Interface Requirements

4.1. User Interfaces Overview

The system will feature a clean, responsive GUI designed with the TUL branding guidelines, utilizing buttons and forms for data entry. Users will interact with the system primarily through a web dashboard that provides visual status indicators and clear error messaging.

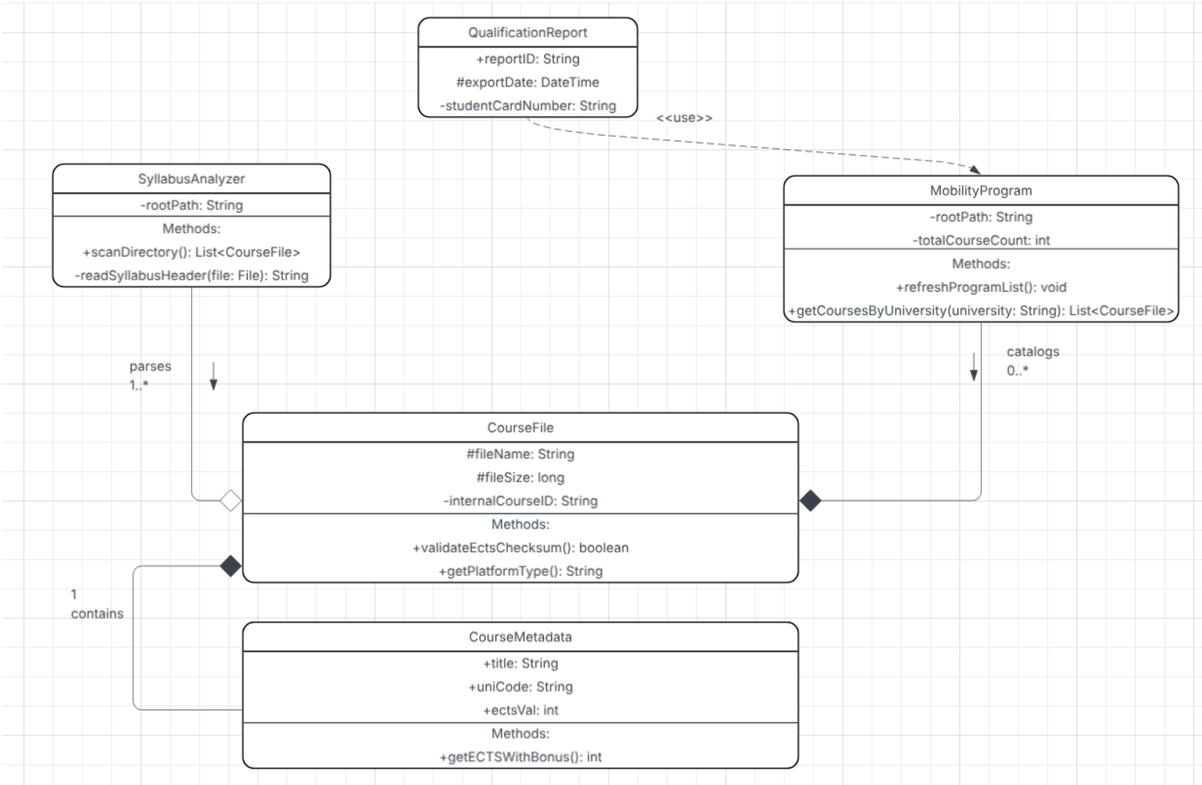
4.2. Hardware Interfaces

The software will interact with standard server hardware and end-user devices like PCs and tablets via standard HTTP/HTTPS protocols. No specialized hardware is required, as the system relies on standard input/output devices for data management.

4.3. Software Interfaces

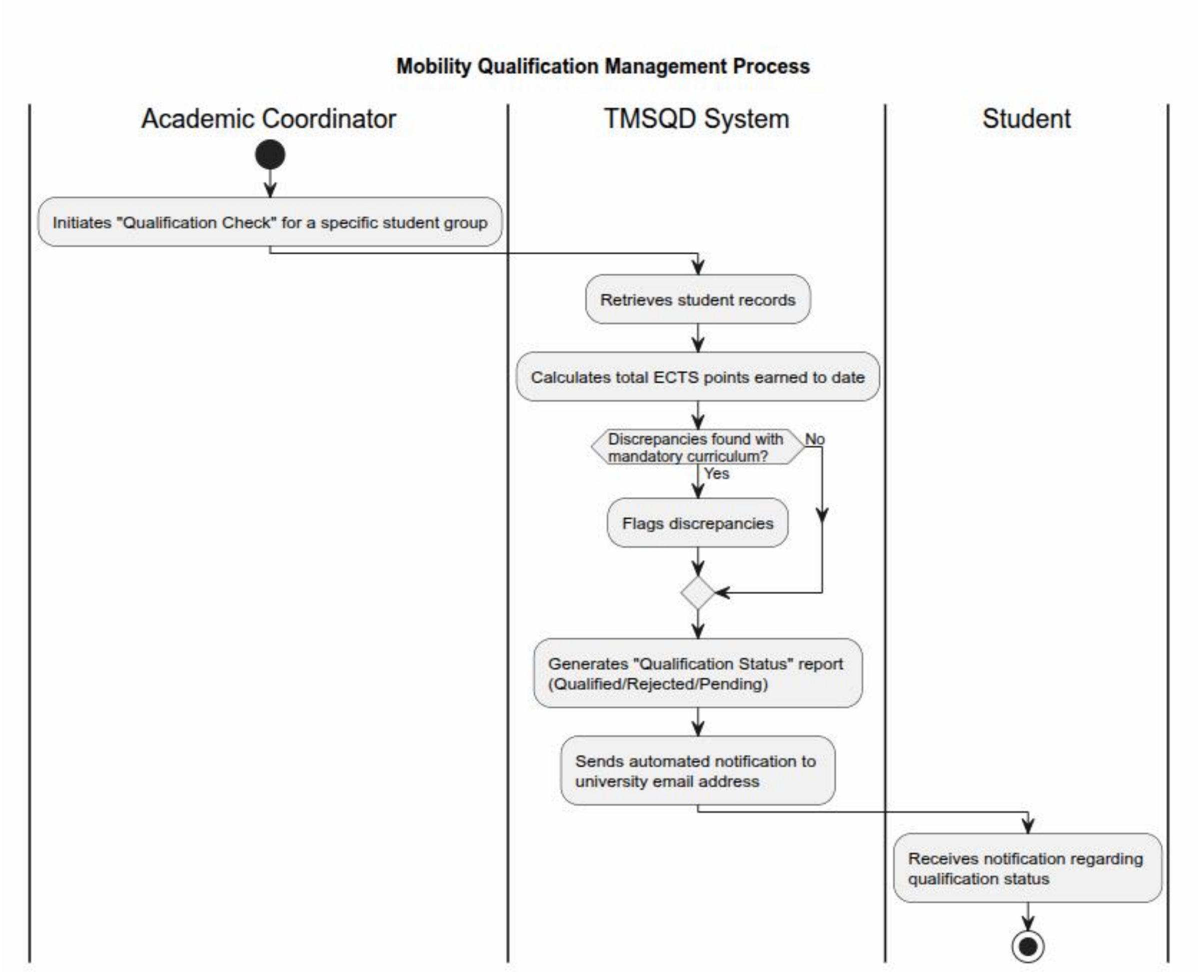
TMSQD will interface with the university's SQL database and the OAuth 2.0 authentication service for secure user login. Data will be exchanged via JSON-formatted messages to ensure seamless integration with other university software components.

5. System Features/Modules



QualificationReport
+reportID: String
#exportDate: DateTime
-studentCardNumber: String
<<use>>
SyllabusAnalyzer
-rootPath: String
Methods:
+scanDirectory(): List<CourseFile>
-readSyllabusHeader(file: File): String
MobilityProgram
-rootPath: String
-totalCourseCount: int
Methods:
+refreshProgramList(): void
+getCoursesByUniversity(university: String): List<CourseFile>
parses
1..*
catalogs
0..*
CourseFile
#fileName: String
#fileSize: long
-internalCourseID: String
Methods:
+validateEctsChecksum(): boolean
+getPlatformType(): String
1
contains
CourseMetadata
+title: String
+uniCode: String
+ectsVal: int
Methods:
+getECTSWithBonus(): int

5.1. Mobility Qualification Management



5.1.1 Description and Priority

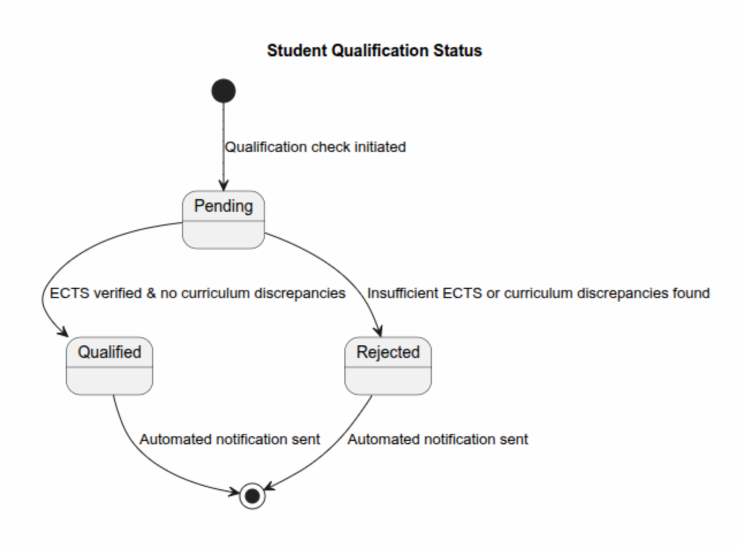
This feature allows for the automated verification of student eligibility for mobility programs based on academic performance and language requirements.

5.1.2 Stimulus/Response Sequences

Stimulus: Coordinator initiates a "Qualification Check" for a specific student group.
Response: The system compares student records against the predefined criteria for the semester.

5.1.3 Functional Requirements

ID	Requirement Description	Priority
REQ-1.1	Upon selection of a student record, the system shall automatically calculate the total ECTS points earned to date.	High
REQ-1.2	The system shall flag any discrepancies between the student's chosen courses and the mandatory curriculum of the mobility semester.	High
REQ-1.3	The system shall provide a "Qualification Status" report (Qualified/Rejected/Pending) for each applicant.	Medium
REQ-1.4	Upon status update, the system shall send an automated notification to the student's university email address.	Medium



5.2.1 Course Quality Verification

Description: A module to store and verify syllabi and learning outcomes of foreign partner universities.

5.2.2 Stimulus/Response Sequence

Stimulus: An Academic Coordinator uploads a syllabus from a partner university and selects the "Verify Quality" option.
Response: The system parses the document, compares ECTS points and learning outcomes against internal TUL standards, and displays a compatibility score.

Stimulus: A user enters the name of a specific foreign course into the mapping search bar.
Response: The system queries the database for historical equivalency records and returns a list of previously approved mappings or a "No records found" notification.

5.2.3 Functional Requirements

ID	Requirement Description	Priority
REQ-2.1	Upon document upload, the system shall scan the file for keywords related to learning outcomes and ECTS values.	High
REQ-2.2	The system shall allow the coordinator to manually confirm or override the automated compatibility score.	Medium
REQ-2.3	Upon a search query, the system shall display all past decisions regarding the specific course to ensure administrative consistency.	High

6. Nonfunctional Requirements

6.1. Performance

- **NF-1.1:** The system shall process qualification reports for up to 500 students in less than 10 seconds.
- **NF-1.2:** Database queries for course searches must return results within 2 seconds under normal load.

6.2. Security

- **NF-2.1:** Access to the database shall be restricted via OAuth 2.0 university credentials.
- **NF-2.2:** Personal student data must be encrypted at rest following GDPR (RODO) standards.
- **NF-2.3:** The system shall maintain an audit log of all manual changes made to qualification statuses.

6.3 Availability

- **NF-3.1:** The system shall maintain 99.5% uptime during the peak qualification periods (April–June).